

Article

Prediction of Students' Adaptability Using Explainable AI in Educational Machine Learning Models

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Abstract: As the educational landscape evolves, understanding and fostering student adaptability has become increasingly critical. This study presents a comparative analysis of XAI techniques to interpret machine learning models aimed at classifying student adaptability levels. Leveraging a robust dataset of 1205 instances, we employed several machine learning algorithms with a particular focus on Random Forest, which demonstrated highest accuracy at 91%. The models' precision, recall and F1-score were also evaluated, with Random Forest achieving a precision of 0.93, a recall of 0.94, and an F1-score of 0.94. Our study utilizes SHAP, LIME, Anchors, ALE, and Counterfactual explanations to reveal the specific contributions of various features impacting adaptability predictions. SHAP values highlighted 'Class Duration' significance (mean SHAP value: 0.175); LIME explained socio-economic and institutional factors' intricate influence. Anchors provided high-confidence rule-based explanations (confidence: 97.32%), emphasizing demographic characteristics. ALE analysis underscored the importance of 'Financial Condition' with a positive slope, while Counterfactual scenarios highlighted the impact of slight feature variations of 0.5 change in 'Class Duration'. Consistently, 'Class Duration' and 'Financial Condition' emerge as key factors, while the study also underscores the subtle effects of 'Institution Type' and 'Load-shedding'. This multi-faceted interpretability approach bridges the gap between machine learning performance and educational relevance, presenting a model that not only predicts but also explains the dynamic factors influencing student adaptability. The synthesized insights advocate for educational policies accommodating socioeconomic factors, instructional time, and infrastructure stability to enhance student adaptability. The implications extend to informed and personalized educational interventions, fostering an adaptable learning environment. This methodical research contributes to responsible AI application in education, promoting predictive and interpretable models for equitable and effective educational strategies.

Keywords: comparative analysis; educational data mining; educational predictive modelling; XAI; feature importance; machine learning interpretability; model transparency; predictive analytics in education; student adaptability; AI in education policy



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1. Introduction

A critical area of concern for educators, policymakers, and academics in the quickly changing field of education is comprehending and improving student adaptation [1,2]. The ability of students to effectively navigate and respond to diverse learning environments, technological advancements, and varying educational methodologies highlights the importance of adaptive learning systems tailored to individual needs [3].

The field of education has seen a revolutionary change with the introduction of machine learning techniques, which have provided new ways to forecast and enhance

student adaptation [4]. However, the black-box nature of complex machine learning models often impedes their interpretability, raising concerns regarding their applicability in educational settings where transparency and comprehensibility are paramount. The inability of educators to understand the machine learning algorithms' decision-making process due to their black-box nature has been a key barrier in the learning space [5,6]. Because of this, there is increasing interest in building and evaluating interpretable machine learning models that can be used to identify student adaptability factors. These models should not only produce precise predictions but also offer valuable insights into the decision-making process.

Student adaptability is a complex construct encompassing the ability to effectively adapt, learn, and succeed in diverse educational settings. It involves factors such as socio-economic background, individual learning styles, technological proficiency, and environmental influences [7,8]. Understanding and measuring adaptability require a holistic approach that integrates diverse data sources and utilizes sophisticated models capable of capturing intricate patterns within these datasets. Even though machine learning models perform predictably well across a range of areas, their opacity poses a substantial difficulty when used in educational settings. Interpretability issues make it more difficult for stakeholders to understand model predictions and restrict the discovery of useful information that is essential for creating customized interventions that improve student flexibility. In order to close this gap, new developments in interpretable machine learning methods provide ways to decipher how complicated models make decisions [9–11].

In response to the evolving challenges and opportunities in education, this study innovates by introducing a comparative analysis of a novel combination of interpretable machine learning models tailored to enhancing our understanding of student adaptability. Through the meticulous employment of SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), Anchor, Accumulated Local Effects (ALE), and Counterfactual explanations, we navigate the intricacies of balancing model interpretability against predictive accuracy. By leveraging a dataset rich in student attributes and adaptability levels, our research offers unprecedented insights into the predictors of adaptability, navigating the complexities inherent in educational data. Grounded in cutting-edge techniques and established educational theories, our methodology facilitates a detailed examination of the myriad factors influencing student outcomes. The application of machine learning in this context, with its capacity to analyze complex datasets and reveal intricate patterns, marks a significant advancement in educational research. This study not only underscores the predictive power and interpretability of machine learning models but also highlights their potential to inform targeted, effective educational interventions, thereby pioneering a comprehensive exploration aimed at unveiling and acting upon the factors influencing student adaptability, thereby fostering informed and personalized educational strategies.

The rest of the paper is arranged as follows: Section 2 provides an overview of related work on XAI and interpretable models for student adaptability. Section 3 describes the methodology. Section 4 presents the results of the experiments, while Section 5 discusses comparative analysis of XAI techniques and synthesis of findings. Finally, Section 6 discusses future work and conclusions.

2. Related Work

Numerous studies have explored the correlation between student adaptability and academic performance, retention rates, and overall learning outcomes. The link between learning flexibility and self-regulated learning (SRL) in junior high school students is examined in the study by She et al. [12], with an emphasis on the mediating roles of academic motivation and self-management. Under China's "double reduction" educational program, the research looks at how these elements work together to support SRL, which is an important ability for students to succeed. The study's conclusions show the independent and cumulative mediation functions of academic motivation and self-management, as well

as the noteworthy benefits of learning adaptability on SRL. This offers information to help students improve their learning practices and adjust to changes in schooling.

Martin et al.'s systematic review [13] summarizes research on adaptive learning from 2009 to 2018, with an emphasis on publishing patterns, instructional settings, techniques, and technology. After analyzing 61 articles, the study found that the number of publications peaked in 2015 and that higher education particularly in Taiwan and the US was the main topic of interest. The study focuses on several stages of instructional design in the field of adaptive learning research, such as learner characteristics, needs assessment, and system design and implementation. Furthermore, it classifies adaptive technologies into systems, applications, teaching techniques, and design solutions, and adaptive strategies into categories based on learner models, content, and instructional models. In the review, it is emphasized how important it is to have more diverse research and rigorous methods in adaptive learning.

In their investigation of student adaptation following COVID-19-induced online learning, Tiwari et al. [14] explore Education 4.0. By using Explainable Machine Learning (XML), they investigate students' acceptance of Industry 4.0 features. Using a range of machine learning techniques, the study finds that Neural Networks are the most accurate (93%) at classifying adaptation. The study emphasizes how crucial explainability is for machine learning models. It uses dataset analysis, model performance research, and SHAP techniques to clarify feature importance and model predictions. In the end, it emphasizes how important elements affecting students' capacity to adjust include age, socioeconomic situation, and type of internet access. This thorough study helps teachers comprehend how pupils adjust to online learning and what effects it has.

Adnan et al. [15] present an XAI model that predicts student performance in virtual learning environments at various course stages with detailed interpretability, utilizing Random Forest among other algorithms and incorporating SHAP values for insight into decision-making processes. This enhances educational data mining by enabling early performance prediction. In a related context, Gligorea et al. [16] develop an interpretable machine learning framework to analyze and predict student performance, focusing on commitment and grades, and improving framework efficiency through algorithmic tuning and GPU-based infrastructure. Another study [5] explores early prediction of student performance using machine learning and XAI. Logistic regression yielded the best results, and the integration of XAI helped educators understand the predictions, aiming to support at-risk students promptly.

The adaptability of Bangladeshi pupils to online learning amid the COVID-19 pandemic is investigated by Suzan et al. [1]. The study uses machine learning models including Decision Tree, Random Forest, Naive Bayes, SVM, KNN, and ANN and conducts surveys across different educational levels. The Random Forest Classifier, which focuses on sociodemographic characteristics, has the best accuracy (89.63%). This study provides insightful information for educational policymakers by illuminating the difficulties and degrees of adaptation faced by students in the context of online learning. The results highlight the importance of comprehending student flexibility for the successful implementation of online learning and offer directions for future study in evaluating sociodemographic impacts on students' mental health in online learning.

This study builds on previous work by Suzan et al. [1], who also utilized the same dataset to predict student adaptability levels during the pandemic. Our research extends this work by applying a comprehensive suite of Explainable AI techniques to provide deeper insights into the factors influencing adaptability. We reference their findings to contextualize our results and highlight the unique contributions of our study.

LIME was introduced by Ribeiro et al. [9]. LIME increases user trust in machine learning models, which are sometimes viewed as mysterious black boxes, by providing clear explanations for every classifier's predictions. By building an interpretable model around the prediction, the method offers faithful and understandable local explanations. Additionally, it suggests a way to use a group of representative individual forecasts to

explain the global model. The study illustrates how LIME may be used to explain a wide range of models and how useful it is in situations when trust is needed, such as when choosing models, enhancing classifiers, and comprehending model choices. Rao et al.'s [17] study explores XAI in healthcare predictive models, highlighting its potential to enhance trust, transparency, and clinical decision-making in medical diagnostics.

A unique model-agnostic method called Anchors is introduced in the publication “Anchors: High-Precision Model-Agnostic Explanations” by Ribeiro et al. [11]. Anchors are used to explain complicated model behaviors using high-precision rules. Anchors give clarity and simplicity of comprehension by representing local variables necessary for predictions. With high-probability assurances, the authors suggest an algorithm to calculate these explanations for every given black-box model. The versatility of Anchors is shown in a number of domains, such as text, tabular data, and pictures. User research demonstrates that, in comparison to current linear explanations, Anchors enable more precise and effective prediction of model behavior on unseen instances. This method offers a compromise between accuracy and understandability in model explanations, addressing the interpretability problem in machine learning.

To examine several approaches to machine learning, Counterfactual explanations have been utilized, highlighting their significance for comprehending and assessing AI judgments. Using determinantal point processes, Mothilal et al.'s research [18] focuses on producing a variety of realistic Counterfactuals while taking user context into account for realism. Their tests demonstrate that their approach generates locally correct Counterfactuals, improving machine learning systems' decision-making capabilities. Van Looveren and Klaise [19] provide a fast, model-independent technique for creating Counterfactual scenarios with class prototypes that works with a variety of data types. By using this approach, black-box model interpretability is enhanced and computational difficulties are resolved. Finally, Wachter et al. [20] address Counterfactual explanations in relation to the General Data Protection Regulation (GDPR), emphasizing how they can make automated judgments more transparent without jeopardizing the specifics of private algorithms. Collectively, these works highlight the significance of Counterfactual explanations in making AI more understandable and accountable.

3. Methodology

In this section, we describe the methodology used in this research, detailing the methods involved in selecting the dataset, preparing the data, and using machine learning models to predict the adaptability levels of students. In order to understand the models and offer insights about feature relevance, we explain the deployment of several Explainable AI (XAI) strategies. The technique aims to bridge the gap between interpretability and predicted accuracy by rigorously testing, evaluating, and training the models. This ensures a thorough examination.

3.1. Dataset

This research dataset is taken from www.kaggle.com (accessed on 18 November 2023) [21] and consists of an instance with 1205 rows and 14 features (columns) with these columns: Gender, Age, Education Level, Institution Type, IT Student, Location, Load-shedding, Financial Condition, Internet Type, Network Type, Class Duration, Self Lms, Device, and the Adaptivity Level which is the target feature with the adaptivity level set to Low, Moderate or High. This dataset was chosen for its rich representation of student demographics, allowing a detailed exploration of student adaptability's multifaceted nature, highlighting the interplay of socio-economic, institutional, and personal factors impacting student adaptability. The dataset was collected during pandemic restrictions, providing a unique context for analyzing student adaptability under extreme conditions. The pandemic significantly impacted educational environments, making this period critical for understanding how students adapt to sudden changes and remote learning conditions. Insights gained from this dataset can offer valuable lessons for future educational disruptions.

In order to ensure a thorough assessment of the prediction performance and interpretability of the machine learning models used, we used 80% of the dataset (964 instances) for training the models and the remaining 20% (241 instances) for testing the models for this study. We trained and tested our machine learning models using an 80/20 split in this study's methodology. The choice was taken in light of our dataset's size and characteristics, which limited the possibility of assigning a distinct validation set without appreciably decreasing the size of the training set. In order to guarantee the accuracy of our model predictions, we placed a high priority on keeping a solid training dataset. We do identify this as a research drawback, though, since an extra layer of model evaluation and fine-tuning may have been supplied by a second validation step.

3.2. Research Framework

This paper's main contribution is its innovative approach to evaluating student adaptability predictions from ML models, a critical component in educational settings. The proposed method, outlined in Figure 1, is a comprehensive, multi-stage process that begins with meticulous data collection, ensuring a robust foundation for subsequent analysis. The preprocessing stage addresses the typical challenges of educational data, such as encoding categorical variables and normalizing features, to prepare a clean, analysis-ready dataset.

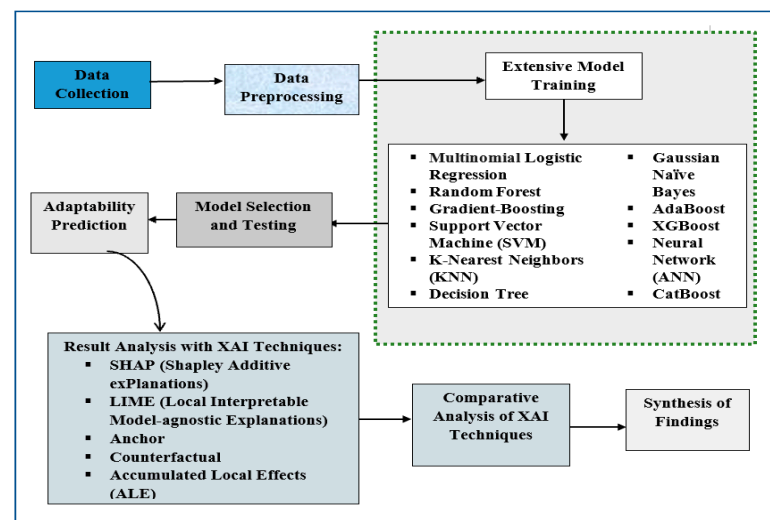


Figure 1. Block diagram for implementing the proposed framework.

The method's thorough training of a wide range of machine learning models, from simpler ones like Multinomial Logistic Regression to more complex ensembles, is one of its distinguishing features. This inclusive approach not only compares predictive power across various algorithms but also enhances the process of interpretation analysis by providing diverse perspectives on the same dataset. The central tenet of the methodology is the application of multiple XAI techniques, such as SHAP, LIME, Anchors, Counterfactuals, and ALE. This diverse mix allows for a detailed exploration of how different features influence the adaptability predictions, providing global and local interpretability. Each XAI technique offers unique insights: SHAP quantifies feature contributions, LIME provides locally faithful explanations, Anchors generate high-precision rules, Counterfactuals suggest minimal changes for outcome alteration, and ALE analyzes feature effects, averting confounding interactions.

The subsequent comparative analysis of these XAI techniques is a methodological innovation that critiques their coherence, identifies consistencies, and pinpoints anomalies in their explanations. The final synthesis of findings distills a holistic understanding of the ML models, offering actionable insights for educational stakeholders. This methodology empowers educators and policymakers with a deeper comprehension of the pre-

dictive models, facilitating informed decisions to foster student adaptability in diverse learning environments.

3.3. Data Preprocessing

We process the data by loading the dataset using the data manipulation library ‘Pandas’ in Python version 3.11.5. Understanding the dataset’s structure is necessary for further investigation. The characteristics and target columns may be identified by reviewing the column titles; to understand the data distributions depicted in Table 1, summary statistics for numerical columns must be collected. Categorical variables are converted into a numerical format using One-Hot Encoding. Scaling of numerical features using the ‘StandardScaler’ method ensures uniformity in the magnitude of features. The mapping for our target variable was Moderate: 2, Low: 1, High: 0. The handling of unbalanced data is then considered and customized to meet the needs of the model and particular dataset features. The dataset is then split into training and testing sets using ‘train_test_split’ for building the model and model evaluation.

Table 1. Summary statistics of the dataset.

Variable	Count	Unique	Top	Freq
Gender	1205	2	Boy	603
Age	1205	6	21–25	374
Education Level	1205	3	School	530
Institution Type	1205	2	Non-Government	623
IT Student	1205	2	No	501
Location	1205	2	Yes	135
Load-shedding	1200	2	Low	1004
Financial Condition	1206	3	Mid	676
Internet Type	1205	2	Mobile	655
Network Type	1205	3	4G	775
Class Duration	1205	3	1–3	840
Self Lms	1205	2	No	955
Device	1205	3	Mobile	1013
Adaptivity Level	1205	3	Moderate	625

Adaptivity Level is the target variable for classification. This column contains three different levels: Moderate, Low and High. Feature distribution of the adaptivity level in the dataset yields the following frequency counts: Moderate: 625, Low: 480, and High: 100; from these counts, you can see that the dataset is not perfectly balanced.

The student adaptivity levels are shown in Figure 2 as a balanced frequency chart after using the Synthetic Minority Over-sampling Technique (SMOTE). This visualization represents the distribution of adaptivity levels (Moderate, Low, and High) after addressing the initial class imbalance inherent in the dataset. Before the application of SMOTE, the adaptivity levels were not proportionately distributed, which could have led to a predictive model biased toward the majority class. By synthesizing new examples of the minority classes, SMOTE equalizes the representation of each adaptivity level, as depicted in the figure.

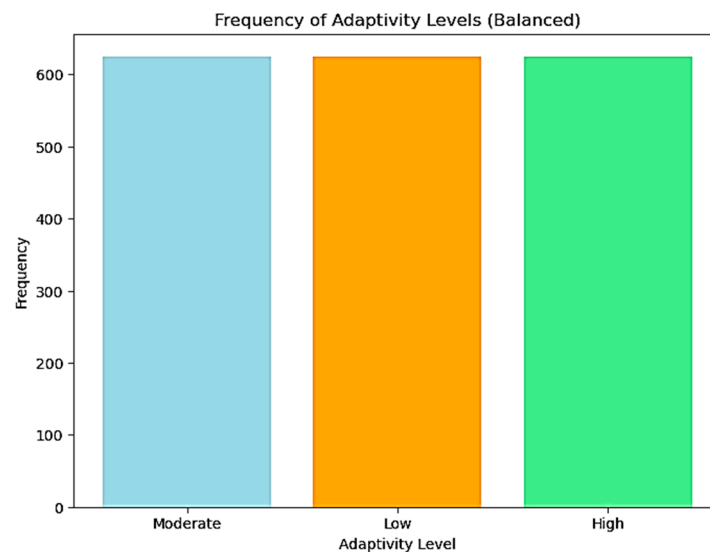


Figure 2. Balanced frequency plot of Adaptivity Level.

Each bar in the graph corresponds to one of the three adaptivity levels, and the height reflects the frequency of instances within the dataset after balancing. The near-equal heights of the bars indicate that each class now has a similar number of instances (625), thus ensuring that the predictive models trained on this dataset are not biased towards any particular adaptivity level.

3.4. Model Training

We undertake a series of extensive classification experiments by applying several machine learning algorithms to classify levels of student adaptability. They were analyzed and classified as shown in Table 2.

Table 2. Models trained and their parameters.

S/N	Model	Parameters	Values
1	Multinomial Logistic Regression	MaximumIterations(max_iter) multi_class	1000 auto
2	Decision Tree	max_depth min_samples_leaf min_samples_split criterion splitter max_features	range (1, 20, 2) range (1, 100, 5) range (2, 10) gini and entropy best and random sqrt
3	Random Forest	random_state	0
4	K-Nearest Neighbors	n_neighbors	5
5	XGBoost	max_depth learning_rate n_estimators	5 0.01 200
6	Gradient Boosting	n_estimators random_state	100 42
7	Gaussian Naive Bayes	Gaussian Naive Bayes	Normal no special setup
8	Support Vector Machine (SVM)	Kernel Regularization Parameter (C) Random State	Linear 1.0 42

Table 2. Cont.

S/N	Model	Parameters	Values
9	Ensemble	Random Forest Classifier	
		random_state	42
		Gradient Boosting Classifier	
		random_state	42
		Logistic Regression	
		max_iter	1500
10	AdaBoost	random_state	42
		Ensemble method used	voting
		Base Classifier	Decision Tree
		max_depth	1
		n_estimators	50
11	CatBoost	random_state	42
		Iterations	1500
		Depth	6
		learning_rate	0.1
		verbose	200
		random_state	42
12	Neural Network	early_stopping_rounds	50
		Input Layer	X_train_stand
		Activation Function	ReLU
		First Hidden Layer:	
		Units	64
		Second Hidden Layer	
		Units	32
		Third Hidden Layer	
		Units	16
		Output Layer	
		Units	3
		Activation Function	Softmax

3.5. Explainable Artificial Intelligence (XAI) Used

The classifications made by the models are then analyzed using several XAI techniques to interpret the model's decision-making process. These techniques include:

3.5.1. SHAP (Shapley Additive exPlanations)

SHAP is a high-level machine learning interpretability method built on game theory concepts to explain the output of any machine learning technique. It was developed by Lundberg and Lee as a unifying approach to explaining the predictions of machine learning algorithms. The SHAP value measures the predictive influence of each feature on the base prediction. The base prediction is typically the average prediction of the dataset used to train the model [10]. The SHAP value for the i -th feature is calculated using the formula:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [F_x(S \cup \{i\}) - f_x(S)] \quad (1)$$

where ϕ_i is the SHAP value for the i -th feature, S is a subset of all features F excluding the i -th feature, $f_x(S)$ is the prediction made by the model using only the features in set S , $|S|$ is the number of features in set S , and $|F|$ is the total number of features.

For example, consider a machine learning model that predicts student adaptability based on various features. Let us calculate the SHAP value for the feature 'Class Duration'. The base prediction (the average prediction of the training dataset) might have a 50% chance of a student being adaptable. If a particular student studies 5 h a day and this changes the prediction to 70% adaptability, the SHAP value for 'Class Duration' in this case would quantify how much of that 20% increase is attributed to studying 5 h specifically,

compared to the average student. To apply this, one would calculate the difference in prediction with and without the ‘Class Duration’ feature over all possible combinations of other features, weighted by the number of features included. This exhaustive calculation considers all possible interactions with other features, thus allocating the ‘Class Duration’ feature its fair share of influence on the increase in adaptability prediction.

The SHAP framework meets several desirable qualities, including consistency, missingness, and local accuracy, and it is model-agnostic, so any machine learning technique can adopt it [22–25]. These characteristics guarantee the accuracy and dependability of SHAP explanations, which makes them especially useful in critical situations where interpretability is just as crucial as predictive performance. SHAP is used in many fields, such as energy to predict power consumption trends, healthcare to assess patient risk, and finance to evaluate credit risk. Its broad use highlights how crucial interpretability is to machine learning [24,26].

The practical application of SHAP in our study allows us to identify the most influential factors determining student adaptability. For example, SHAP analysis revealed that “Class Duration” and “Financial Condition” are critical in predicting adaptability levels. This insight is invaluable for educational institutions seeking to understand the determinants of student success and can inform targeted interventions to support student adaptability.

3.5.2. LIME (Local Interpretable Model-Agnostic Explanations)

LIME is a machine learning technique that provides an accurate and understandable explanation for the classification made by machine learning models [9]. LIME works by perturbing input data and observing changes in the model’s predictions. It then selects the most influential features for a particular prediction and fits a simple model, such as a linear regression, to these features. LIME seeks to explain the reasoning behind a model’s prediction y for an individual instance x by approximating the complex model f locally with an interpretable model g . This interpretable model, typically a linear model, operates on a binary feature representation z' , which is derived from the original feature space of x . The local surrogate model is defined as:

$$g(z') = \beta_0 + \sum_{i=1}^n \beta_i z'_i \quad (2)$$

where $g(z')$ represents the prediction of the interpretable model, β_0 is the intercept term of the linear model, β_i represents the weight associated with the i -th feature in the interpretable space, and z'_i is the binary representation of the i -th feature.

LIME aims to optimize the following objective function:

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} \mathcal{L}(f, g, \pi_x) + \Omega(g) \quad (3)$$

where $\xi(x)$ is the local explanation for the instance x , $\mathcal{L}$ measures the fidelity of g in approximating f within the locality defined by π_x , $\Omega(g)$ quantifies the complexity of the interpretable model g , and G denotes the family of interpretable models, such as linear regressions or decision trees.

Consider a machine learning model that predicts a student’s adaptability level. For a particular student, the model predicts a “High” adaptability level. LIME is employed to explain this prediction, focusing on features like ‘Class Duration’, ‘Financial Condition’, and ‘Institution Type’, etc. Perturbing the input data around the student’s features, LIME fits a local linear model to these perturbations to approximate the prediction of the original complex model f . The linear model constructed by LIME provides the following coefficients for an interpretable explanation:

Adaptability Score = $0.10 + 0.25 \times \text{Class Duration} - 0.15 \times \text{Financial Condition} + 0.10 \times \text{Institution Type}$. An increase in ‘Class Duration’ by one unit (e.g., one hour) enhances the adaptability score by 0.25, suggesting the critical role of engagement time in

educational adaptability. A decline in ‘Financial Condition’ from ‘Mid’ to ‘Poor’ (encoded as a decrease by one unit) reduces the adaptability score by 0.15, highlighting the potential barriers faced by economically disadvantaged students. A positive change in ‘Institution Type’ from ‘Government’ to ‘Non-Government’ (encoded as an increase from 0 to 1) raises the adaptability score by 0.10, possibly reflecting the influence of institutional resources and support.

LIME explanation shows how each feature impacts the adaptability prediction. The positive coefficient for ‘Class Duration’ underscores the importance of classroom engagement in fostering adaptability, suggesting that longer classes may lead to better student outcomes. The negative coefficient for ‘Financial Condition’ points to the need for targeted support to students from less wealthy backgrounds. The positive coefficient for ‘Non-Government’ institutions indicates potential benefits of these environments, which may include innovative teaching methods or more personalized attention.

Educational policymakers can leverage these insights to ensure that class time is used effectively, provide additional resources for students in need, and encourage beneficial practices observed in ‘Non-Government’ institutions. This approach proves especially valuable in domains where decisions are individual-specific, such as healthcare, education, and finance [27]. The localized interpretability provided by LIME empowers stakeholders to make informed decisions, promoting adaptability and improving outcomes in these critical areas.

3.5.3. Anchor

The Anchor algorithm is designed to provide high-precision rules, known as anchors, which offer interpretable and reliable explanations for the decisions of machine learning models [8]. The interpretable explanations are presented as high-precision rules, which are local conditions sufficient to guarantee the prediction with high confidence. An anchor explanation for a prediction can be expressed as:

$$\text{IF}(A_1 \wedge A_2 \wedge \dots \wedge A_n) \text{ THEN prediction } P \text{ with confidence } c \quad (4)$$

where A_i are anchor conditions (predicates) for features, P is the predicted class or outcome, and c is the confidence level associated with the rule. The algorithm aims to maximize the precision (confidence) of the anchor while ensuring that the coverage (the proportion of instances for which the anchor applies) is above a user-defined threshold. The objective function can be formulated as:

$$\text{Maximize Precision}(A) \text{ subject to Coverage}(A) > \theta \quad (5)$$

where $\text{Precision}(A)$ is the probability that the prediction P is correct given that anchor A applies, $\text{Coverage}(A)$ is the proportion of instances in the dataset for which anchor A applies, and θ is the coverage threshold.

Suppose our machine learning model has predicted the adaptability level of students. To provide an interpretable explanation using the Anchor algorithm, consider an instance where the model predicts a student’s adaptability level as “Moderate”. After the application of the Anchor algorithm, the following rule-based explanation is produced:

IF (Network Type > 1) $\wedge$ (Age $\leq$ 1) $\wedge$ (Gender > 0) $\wedge$ (Institution Type > 0) $\wedge$ (Financial Condition $\leq$ 1) THEN Predict ‘Moderate’ with confidence 97.32%.

This explanation infers that for this particular student, the presence of certain conditions such as having access to a 3G or 4G network, being in the ‘16–20’ age group, being female, attending a non-government institution, and not being classified as ‘Rich’ in financial status, are collectively sufficient to predict a ‘Moderate’ adaptability level with high confidence. This rule-based explanation provides clarity on the factors influencing the adaptability prediction. It suggests that access to quality internet and certain demographic characteristics are significant predictors of adaptability. The high confidence level associated with this anchor allows educators to trust the prediction under the specified

conditions. By identifying and understanding these anchor conditions, educational institutions can tailor support and interventions to improve student adaptability. For instance, efforts can be made to enhance internet access for students, and special attention can be given to gender-specific educational experiences and resources for students from different socio-economic backgrounds.

3.5.4. Accumulated Local Effects (ALE)

ALE is a technique used to interpret the predictions of machine learning models, specifically focusing on understanding how features affect predictions on average. ALE addresses the limitation of global methods that average effects over the entire data distribution, which can be misleading if there are interactions between features. This method was developed to provide a better understanding of the feature effects by averaging them locally and thus avoiding the introduction of such interactions, as described by [28].

The ALE of a feature is given as X_j at a particular value x_j , which is calculated by integrating the marginal effects of X_j over the range of the dataset, while marginalizing over the distribution of other features. For a given feature X_j , the ALE is computed as follows:

$$ALE(x_j) = \int_{x_{j,min}}^{x_j} \left(E \left[\frac{\partial f(x)}{\partial x_j} | X_j = z \right] - E \left[\frac{\partial f(x)}{\partial x_j} \right] \right) dz \quad (6)$$

where $f(x)$ is the prediction model, $\frac{\partial f(x)}{\partial x_j}$ is the partial derivative of the model prediction with respect to the feature X_j , $x_{j,min}$ is the minimum value of the feature X_j in the dataset, and $E[\cdot | X_j = z]$ is the conditional expectation over the distribution of the dataset given the feature $X_j = z$.

Consider a dataset where we aim to understand the impact of ‘Class Duration’ on the adaptability level of students. We can calculate the ALE for ‘Class Duration’ by following these steps:

1. Discretize ‘Class Duration’ into a grid of values.
2. For each interval in the grid, compute the difference in the model’s prediction as ‘Class Duration’ changes within that interval.
3. Accumulate these local effects across the grid to estimate the average effect of ‘Class Duration’ on the model’s prediction.

Presume after computing the ALE, we find that increasing ‘Class Duration’ from 1 h to 3 h increases the model’s prediction for ‘High’ adaptability by 0.2 on average. This implies that longer class durations are associated with a higher likelihood of students being classified as highly adaptable.

The ALE plot for ‘Class Duration’ might show a positive slope, which indicates that as ‘Class Duration’ increases, so does the adaptability prediction. Educational stakeholders can interpret this to mean that policies or interventions designed to extend class durations could potentially improve student adaptability levels. By integrating ALE into the predictive modeling process, educators and policymakers can gain a more detailed understanding of the factors that influence student outcomes. This understanding is crucial for designing educational environments that foster adaptability, a key component of student success in dynamic learning contexts.

3.5.5. Counterfactual Explanations

Counterfactual explanations are interpretability methods in machine learning that focus on “what-if” scenarios, illustrating minimal changes needed for desired output predictions. Originating from the philosophy of causation and decision-making, they provide insights into complex model decision-making, especially in classification tasks. Popularized by [14], they aim to provide explanations for machine learning model decisions that meet legal and non-discrimination requirements.

Counterfactual explanations in machine learning involve finding an instance x' that is minimally different from the original instance x but leads to a different prediction outcome. Counterfactual explanation seeks to solve the following optimization problem:

$$\min_{x'} d(x, x') \text{ subject to } f(x') \neq f(x) \quad (7)$$

where $f(x)$ is the prediction of the original instance, $f(x')$ is the prediction of the Counterfactual instance, and $d(x, x')$ is a distance metric (such as Euclidean distance) that quantifies the dissimilarity between the original instance and the Counterfactual.

For example, imagine a machine learning model predicts a 'Low' adaptability level for a student based on features such as 'Class Duration', 'Financial Condition', and 'Institution Type'. A Counterfactual explanation might reveal that a slight increase in 'Class Duration' could change the prediction to 'High' adaptability.

For instance, if the original feature values are $x = [1$ (Low Class Duration), 0 (Poor Financial Condition), 0 (Government Institution)] and the model's prediction $f(x)$ is 'Low adaptability', a Counterfactual explanation would search for a new set of feature values x' that lead to a different prediction $f(x') = \text{'High adaptability'}$. Suppose the Counterfactual feature values are $x' = [2$ (High Class Duration), $0, 0]$, indicating that only 'Class Duration' needs to be increased to alter the prediction. The Counterfactual explanation therefore might suggest that improving the 'Class Duration' from Low to High could enhance the student's adaptability level according to the model.

Counterfactual insights, like increased instructional time positively affecting student adaptability, guide educational resource allocation and intervention strategies. Educators, leveraging Counterfactual explanations, pinpoint influential factors in the learning process for targeted improvements. This forms a feedback loop, enhancing both educational strategies and machine learning models. Beyond education, Counterfactual explanations facilitate understanding and actionable insights in finance, revealing changes for loan approval, and in healthcare, illustrating shifts in patient risk categories based on lifestyle or clinical parameters [29,30].

4. Results and Discussion

This section presents the results of machine learning models used to predict student adaptability levels. It details the experimental setup, computational environment, and model parameters. The models' performance is analyzed using metrics like accuracy, precision, recall, and F1-score. The interpretability of the models is explored using Explainable AI techniques. Key features' impact on adaptability predictions is highlighted.

4.1. Experimental Setup

A personal computer running Windows 10 Pro, equipped with a Lenovo X1 Yoga (Thinkpad) (Lenovo Group Ltd., Hong Kong, China) Core i7-6600U CPU, 16 GB of DDRAM, and a 256 GB SSD, was used for the research. The Anaconda, Inc.-packaged Python version 3.11.5 was utilized to implement the machine learning models. Context for the computing performance of the methods employed in this study is provided by this hardware specification.

4.2. Model Selection and Prediction

A thorough comparison between many machine learning algorithms was carried out in an effort to identify the best model for the classification of student adaptability levels. These include Multinomial Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), XGBoost, Gradient Boosting, Gaussian Naïve Bayes, Support Vector Machine (SVM), Ensemble methods, AdaBoost, CatBoost, and Neural Networks. Tables 2 and 3 include confusion matrices and a variety of performance metrics, providing a comprehensive summary of each model's predictive performance.

Table 3. Confusion Matrices of the Deployed Models.

	Model	Class Name	Predicted		
			High	Low	Moderate
Actual	Multinomial Logistic Regression	High	11	5	7
		Low	1	57	45
		Moderate	3	14	98
	Decision Tree	High	15	0	8
		Low	2	95	6
		Moderate	1	7	107
	Random Forest	High	15	0	8
		Low	2	97	4
		Moderate	0	7	108
	K-Nearest Neighbors	High	10	1	12
		Low	1	89	13
		Moderate	2	19	94
	XGBoost	High	12	0	11
		Low	3	67	33
		Moderate	2	8	105
	Gradient Boosting	High	13	0	10
		Low	2	87	14
		Moderate	0	14	101
	Gaussian Naive Bayes	High	13	3	7
		Low	2	62	39
		Moderate	6	20	89
	Support Vector Machine (SVM)	High	13	1	9
		Low	3	57	43
		Moderate	4	9	102
	Ensemble	High	12	1	10
		Low	0	92	11
		Moderate	0	4	111
	AdaBoost	High	8	3	12
		Low	2	69	32
		Moderate	3	22	90
	CatBoost	High	15	0	8
		Low	2	95	6
		Moderate	0	8	107
	Neural Network	High	15	0	8
		Low	2	93	8
		Moderate	0	7	108

The three adaptability classes (High, Low, and Moderate) of each model's predictions were broken down in depth by the confusion matrices as shown in Table 3. For example, the Multinomial Logistic Regression model accurately predicted 'High' adaptability 11 times, but it also incorrectly categorized 7 'Moderate' and 5 'Low' instances as 'High'.

Understanding the predictive dynamics of each classifier, including their advantages and disadvantages in differentiating between classes, depends on the specificity of the data.

To give a complete view of each model's effectiveness, the associated performance measures were generated, including accuracy, precision, recall, and F1 score for each class, as shown in Table 4. The Random Forest model had a strong predictive capacity overall, with precision values of 0.93, 0.94, and 0.94 for the High, Low, and Moderate classes, respectively. These results are further supported by the F1 scores, where high values show a balanced precision-recall trade-off.

Table 4. Performance Metrics of The Deployed Models.

Model	Class Name	Precision	Recall	F1-Score
Multinomial Logistic Regression	High	0.73	0.48	0.58
	Low	0.75	0.55	0.64
	Moderate	0.65	0.85	0.74
Decision Tree	High	0.88	0.65	0.75
	Low	0.88	0.92	0.90
	Moderate	0.88	0.89	0.88
Random Forest	High	0.88	0.65	0.75
	Low	0.93	0.94	0.94
	Moderate	0.90	0.94	0.92
K-Nearest Neighbors	High	0.77	0.43	0.56
	Low	0.82	0.86	0.84
	Moderate	0.79	0.82	0.80
XGBoost	High	0.71	0.52	0.60
	Low	0.89	0.65	0.75
	Moderate	0.70	0.91	0.80
Gradient Boosting	High	0.87	0.57	0.68
	Low	0.86	0.84	0.85
	Moderate	0.81	0.88	0.84
Gaussian Naive Bayes	High	0.62	0.57	0.59
	Low	0.73	0.60	0.66
	Moderate	0.66	0.77	0.71
Support Vector Machine (SVM)	High	0.65	0.57	0.60
	Low	0.85	0.55	0.67
	Moderate	0.66	0.89	0.76
Ensemble	High	1.0	0.52	0.69
	Low	0.95	0.89	0.92
	Moderate	0.84	0.97	0.90
AdaBoost	High	0.62	0.35	0.44
	Low	0.73	0.67	0.70
	Moderate	0.67	0.78	0.72

Table 4. Cont.

Model	Class Name	Precision	Recall	F1-Score
CatBoost	High	0.88	0.65	0.75
	Low	0.92	0.92	0.92
	Moderate	0.88	0.93	0.91
Neural Network	High	0.88	0.65	0.75
	Low	0.93	0.90	0.92
	Moderate	0.87	0.94	0.90

A comparison of the accuracy obtained by several machine learning algorithms used in the classification of student adaptability levels is shown in Figure 3. The accuracy of an algorithm is represented by each bar, which is measured as the proportion of test cases that are correctly classified, with Random Forest proving the overall best with 91%, followed by Neural Network and CatBoost with 90% each. Performance metrics and confusion matrices highlight how crucial model selection is in the field of educational data analytics. In particular, the Random Forest, CatBoost, and Neural Network models have demonstrated encouraging possibilities for use in learning environments with the goal of predicting and improving student adaptability.

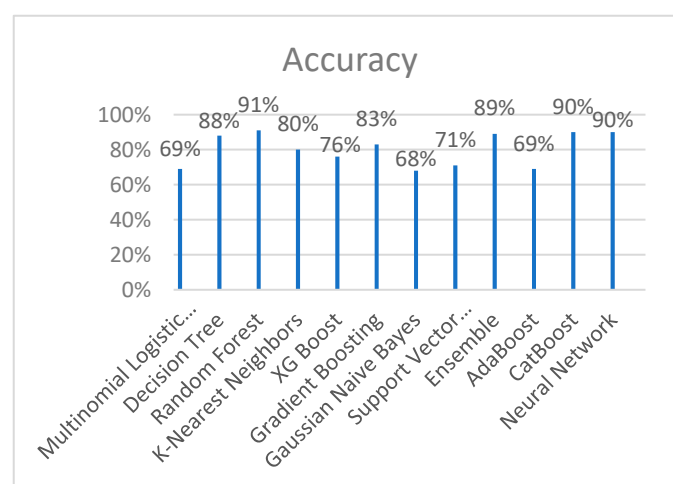


Figure 3. Model accuracy comparison.

4.3. Result Analysis with XAI Techniques

With our range of machine learning models, of which Random Forest performed the best with 91% predictive accuracy, we sought to understand the decision-making processes that buttress the performance of XAI. The following XAI techniques were used to illuminate the contributions and influences of features within our models.

4.3.1. SHAP (Shapley Additive exPlanations)

Using the mean SHAP as a guide, Figure 4 provides a detailed examination of how different factors affect the adaptability levels of students. SHAP values quantify the contribution of each feature to the predictive model's output, offering a measure of importance based on average impact magnitude. This bar graph segregates the influence of features into three adaptability predictions: High (green), Moderate (blue), and Low (pink). The length of each bar represents the mean absolute SHAP value, which is a composite measure of both the strength and consistency of a feature's effect on the model's predictions. 'Class Duration' emerges as a dominant feature, its substantial mean SHAP value indicating a

significant and positive correlation with student adaptability across all levels. Its greatest impact is observed in the High adaptability category, suggesting that extended instructional periods may enhance a student's ability to adapt to varying educational demands.

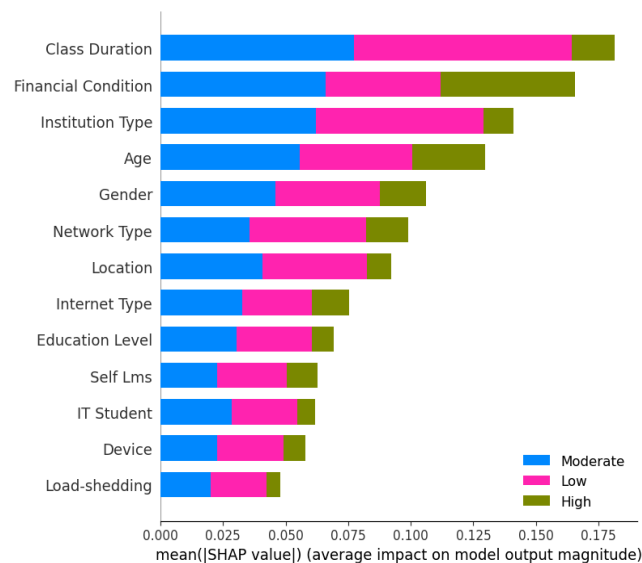


Figure 4. (Global Explanation) Mean impact of features on predicted student adaptability levels using SHAP values.

‘Financial Condition’ is another prominent feature, especially in predicting Low adaptability, highlighting the potential obstacles faced by students with fewer economic resources. Similarly, ‘Institution Type’ displays a varied influence, with a positive association for High adaptability, possibly reflecting the superior resources and support available at certain institutions.

On the other hand, ‘Load-shedding’ predominantly affects Low adaptability predictions, alluding to the detrimental effects of inconsistent electricity on educational continuity. Meanwhile, demographic attributes such as ‘Age’ and ‘Gender’ show moderate effects, indicating their complex but less pronounced roles. Technology-related attributes, namely ‘IT Student’ status and ‘Device’ usage, although impactful, have lesser mean SHAP values, suggesting their influence is secondary compared to educational and socioeconomic factors.

This figure illustration using SHAP values highlights how diverse student adaptation is. It gives educational stakeholders a data-driven platform to build focused interventions that enhance positive aspects and lessen negative ones, creating a climate that encourages flexible learning.

The use of SHAP in a prediction model to ascertain student adaptability levels is seen in Figure 5. The figure is divided into three parts: (a) the attributes under analysis, (b) the corresponding SHAP values, and (c) the force plot. The base value ($f(x)$) represents the average output when no features affect the prediction. In this instance, the model predicted a ‘Low’ adaptability level, indicated by a prediction value of 1. Features with red segments, such as ‘Education Level’ and ‘Network Type’, contribute to a higher adaptability prediction (‘Moderate’ or ‘High’), while features with blue segments, such as ‘Financial Condition’ and ‘Self Lms’, are associated with a lower adaptability prediction (‘Low’). The force plot elucidates the model’s complex reasoning for an individual prediction. For example, a favorable ‘Financial Condition’ and access to ‘Internet Type’ influence the adaptability prediction away from ‘High’, while the absence of engagement with ‘Self Lms’ and lower ‘Education Level’ push the prediction towards ‘Low’ adaptability. The analysis of this figure can help educational institutions understand the complex causes of student adaptability and fit their resources and interventions accordingly.



Figure 5. SHAP force plot for an individual prediction of student adaptability. (a) Instance for an explanation; (b) SHAP value of the instance; (c) instance local explanation.

4.3.2. LIME (Local Interpretable Model-Agnostic Explanations)

A single prediction from a machine learning model that categorizes student adaptability levels is shown in Figure 6, along with LIME study. The adaptation levels that the model predicts are 'Moderate' (probability = 0.62) and 'Low' (probability = 0.38). The prediction for 'High' adaptability is highly unlikely based on the given feature values.

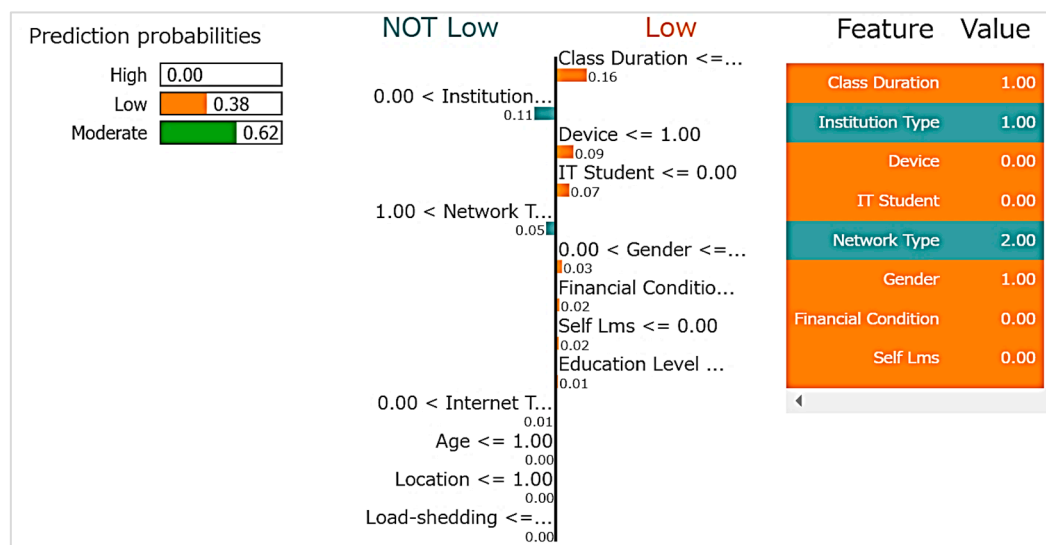


Figure 6. (Local) Interpretation of a model prediction using LIME.

The middle section illustrates the weighted impact of each feature on the prediction decision, with positive contributions towards a 'Not Low' adaptability prediction. For

instance, ‘Class Duration’ and ‘Institution Type’ exhibit the most significant positive influence, suggesting that longer class durations and certain institutional characteristics may be associated with higher adaptability. Conversely, features such as ‘Device’ and ‘IT Student’ status have a notable negative weight, implying that the absence of certain technological factors may lead the model to predict a lower adaptability level.

The rightmost section lists the values of the features for the specific instance being explained, with orange representing positive contributions and blue for negative ones. This visualization aids in understanding the local behavior of the machine learning model, providing transparency into how feature values are aggregated into a predictive outcome. The LIME interpretation offers valuable insights for educators and policymakers by pinpointing specific areas that could be addressed to potentially improve a student’s adaptability. It also emphasizes the intricate nature of the factors that impact educational outcomes and highlights the requirement for sophisticated approaches in educational interventions.

4.3.3. ANCHOR Explanation

Table 5 illustrates how the Anchor algorithm predicts a ‘Moderate’ level of adaptability in students when present, based on specific features. Network Type greater than 1.00 indicates that at least 3G or 4G connectivity is associated with a ‘Moderate’ adaptability prediction, reflecting the importance of reliable internet access for students’ ability to adapt to contemporary educational demands. Age less than or equal to 1.00 corresponds to students in the age group ‘16–20’, revealing that adaptability may be particularly complex during late adolescence. Gender greater than 0.00 suggests that female students are predicted to have ‘Moderate’ adaptability, indicating gender-specific educational experiences or socialization patterns that influence adaptability. Institution Type greater than 0.00 refers to students attending ‘Non-Government’ institutions, suggesting that these institutions may have different resource levels or pedagogical approaches. Financial Condition less than or equal to 1.00 captures ‘Mid’ or ‘Poor’ economic backgrounds, emphasizing the role of socioeconomic status in shaping a student’s adaptability.

Table 5. (Local) Anchor explanation for a ‘Moderate’ adaptability prediction.

Anchor Explanation Condition	
Feature	Condition
Institution Type	>1.00
Age	≤1.00
Gender	>0.00
Financial Condition	≤1.00
Precision	0.9732360097323601
Coverage	0.0888
Predicted Class	Moderate

The precision and coverage metrics accompanying these rules indicate a high level of reliability and relevance within the dataset, suggesting these findings are consistent and pertinent to a subset of the student population. By integrating these interpretive insights into the study, educators and policymakers can better understand the factors contributing to student adaptability levels and inform initiatives to upgrade digital infrastructures. The distinction between government and non-government institutions in adaptability predictions may reflect differential access to resources, student support, and extracurricular opportunities, prompting a re-evaluation of resource distribution to ensure all institutions can foster student adaptability.

4.3.4. Accumulated Local Effects (ALE) Explanation

The ALE feature importance analysis, as visualized in Figure 7, provides compelling insights that can guide educational strategies and policy-making. The significant importance of ‘Class Duration’ suggests that students benefit from extended learning sessions, which may offer a more immersive educational experience. This finding supports the pedagogical approach of increasing instructional time to enhance understanding and retention. Policymakers might consider revising academic schedules to integrate long class periods or more intensive study sessions that could better support students’ adaptability and learning outcomes.

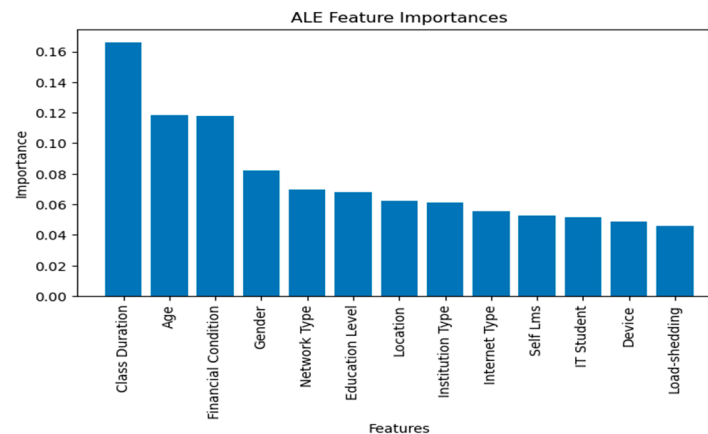


Figure 7. ALE feature importance for predicting student adaptability level.

The importance of ‘Age’ as a predictive factor of adaptability underscores the importance of developmental considerations in educational planning. Younger students may require more structured support to foster adaptability, such as resilience training and social-emotional learning programs. In contrast, older students may benefit from opportunities that challenge their adaptability skills, such as project-based learning and collaborative assignments that mirror real-world scenarios.

The impact of ‘Financial Condition’ on adaptability predictions reinforces the link between economic factors and educational success. This suggests an urgent need for policies that aim to level the playing field, such as providing financial aid, resources, and support systems for students from less affluent backgrounds. Ensuring that all students have equal access to educational resources is not just a matter of fairness but also a strategic investment in the adaptability and resilience of the future workforce.

Gender, network type, and education level highlight the multifaceted nature of adaptability. These findings suggest a tailored approach where educational interventions are sensitive to gender dynamics, technological access, and institutional characteristics. For example, initiatives to bridge the digital divide by improving network connectivity can have far-reaching effects on students’ ability to access and engage with digital learning platforms, a necessity in an increasingly connected world.

Furthermore, the lower importance of features such as ‘Self Lms’, ‘IT Student’, and ‘Device’ does not diminish their value but rather indicates that their influence on adaptability may be more conditional or indirect. This could inform a carefully balanced approach to technology integration in education, ensuring that technology enhances learning without widening the gap between different student groups.

Lastly, the feature ‘Load-shedding’ highlights the external challenges students face, pointing to the broader social and infrastructural issues that can affect educational outcomes. Addressing such challenges may require collaborative efforts that extend beyond the education sector, involving partnerships with community organizations and government agencies to provide stable learning environments.

The ALE feature importance analysis thus serves as a guide for developing comprehensive educational policies and practices that consider the complex interplay of individual, institutional, and societal factors affecting student adaptability. By focusing on these key features, educators and policymakers can create more supportive and effective learning environments that cater to the diverse needs of students, fostering an educational ecosystem that is both equitable and conducive to developing adaptable learners.

Our analysis utilized ALE plots to visualize the impact of three critical features—Class Duration, Age, and Financial Condition—on the predicted adaptability levels of students. The resulting plots offer insightful revelations that intersect with contemporary educational theories and bear significant policy implications, see Figure 8.

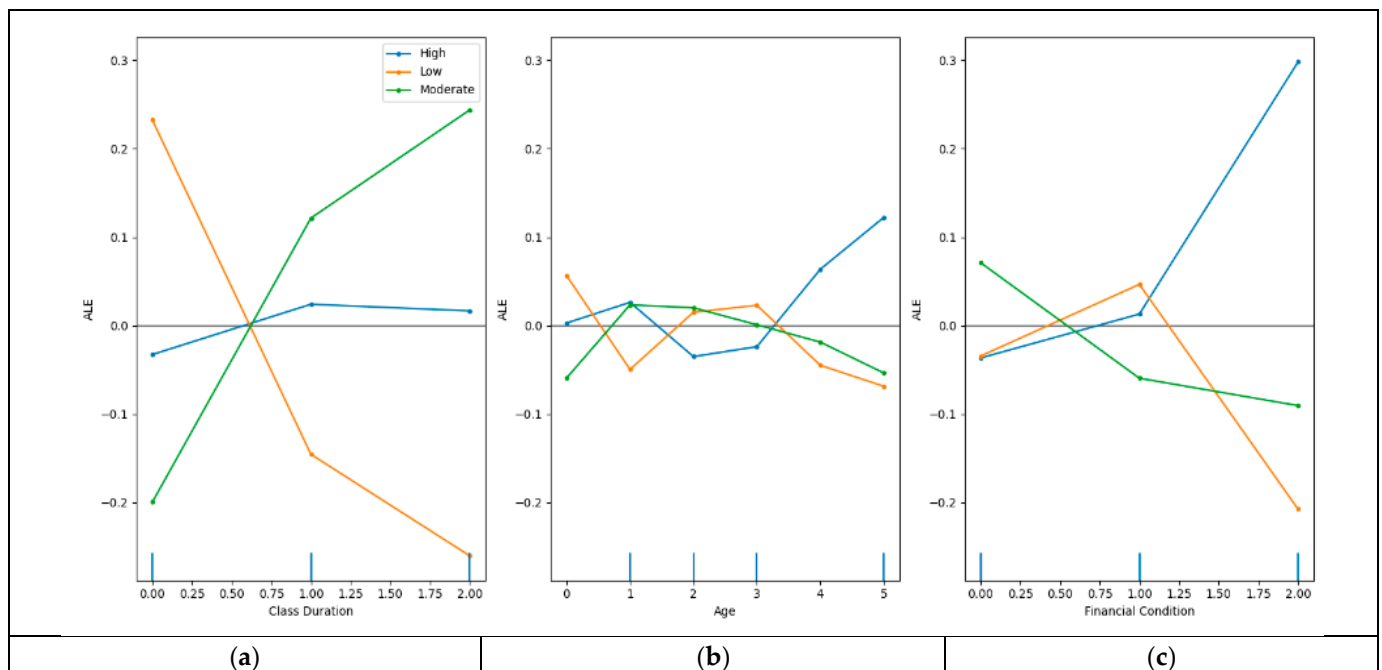


Figure 8. ALE plots for the three most influential features, (a) Class Duration, (b) Age, and (c) Financial Condition, on student adaptability predictions.

Class Duration:

The ALE plot for ‘Class Duration’ indicates that longer class times are positively associated with ‘High’ adaptability predictions. This finding aligns with educational theories that stress the importance of sustained engagement for deeper learning and adaptability. It suggests that extended instructional periods may provide students with more opportunities to assimilate information, engage with challenging concepts, and develop critical thinking skills, all of which are crucial for adaptability in rapidly changing educational landscapes. Policy-wise, this supports arguments for restructuring school schedules to allow for longer class periods, potentially leading to improved educational outcomes.

Age:

The relationship between ‘Age’ and adaptability is less direct, as shown by the minor fluctuations across adaptability levels. However, the general trend indicates that ‘Low’ adaptability decreases with age. This could reflect the development of coping mechanisms and resilience as students mature, a concept supported by developmental theories. The varied adaptability across age groups could inform the design of age-specific curricula and support services, tailoring educational strategies to the developmental stage of the student cohort.

Financial Condition:

The steep positive slope for ‘High’ adaptability with improved ‘Financial Condition’ underscores the pivotal role of economic stability in educational success, as posited by numerous studies linking financial security with better academic performance. The plot highlights a stark reality: students from more affluent backgrounds are likely to be more adaptable, possibly due to greater access to resources, extracurricular activities, and learning support. This insight has profound policy implications, emphasizing the need for equity-focused financial initiatives such as scholarships, grants, and resource allocation to schools serving economically disadvantaged communities.

The ALE plots also reveal the multidimensional nature of student adaptability, validating educational theories, influencing classroom and curriculum design, and emphasizing the need for comprehensive policies to address diverse factors contributing to student adaptability.

4.3.5. Counterfactual Explanation

Figure 9 details a Counterfactual analysis for Instance 3, where a model’s original prediction of a student’s adaptability level as ‘Moderate’ (coded as 2) is altered to ‘Low’ (coded as 1). This analysis is pivotal in understanding the sensitivity of the predictive model to changes in feature values. The original feature vector [1 3 2 1 0 1 1 0 1 1 1 1] encapsulates the student’s profile, with each element corresponding to various features such as Gender, Age, Education Level, etc., as outlined in the research methodology. The Counterfactual vector [1. 3. 2. 1. 0. 1. 1. 0. 1. 1. 0.49932724 1. 1.] presents the minimal adjustment required to flip the prediction outcome. Notably, the alteration occurs at the eleventh feature, which we can infer relates to ‘Class Duration’ given the ‘Feature differences’ array indicating a change of [0.5]. This modification in ‘Class Duration’ decreases its value by 0.5. Considering our encoding scheme, where ‘Class Duration’ might be defined within a range (e.g., ‘0’ for Low, ‘1’ for Moderate, ‘2’ for High), a decrease suggests a shift towards a shorter class duration. The model’s understanding that a reduced ‘Class Duration’ indicates a ‘Low’ adaptability level may stem from a pattern observed in the data, where shorter class durations are linked to low adaptability.

```

Instance 3 - Original pred: 2
Instance 3 - CF pred: 1
Instance 3 - Original: [1 3 2 1 0 1 1 0 1 1 1 1]
Instance 3 - Counterfactual: [1. 3. 2. 1. 0. 1. 1. 0. 1. 1. 0.49932724 1. 1.]
1. 0. 1. 1. 0.49932724 1.
1. ]
Instance 3 - Feature differences: [0.5]

```

Figure 9. Counterfactual analysis illustrating adaptability prediction shift.

The implications of this finding are significant for educational practices. It suggests that ensuring adequate class time may be critical for student adaptability, echoing educational theories that emphasize the role of engagement time in skill development and learning adaptability. Policymakers may need to consider this relationship when designing curriculum schedules, advocating for sufficient instructional time to foster an adaptable learning mindset. Also, this Counterfactual result raises important questions for future research. Why might a slight reduction in ‘Class Duration’ correlate with a lower adaptability prediction? Does this relationship hold across various subjects and learning contexts? Such questions highlight the need for a deeper examination of the educational factors that contribute to student adaptability and the development of interventions that can effectively support students’ learning journeys.

The comparison between the original and Counterfactual feature values for Instance 3 is shown in Figure 10. This comparison offers valuable insight into how the model predicts

student adaptability levels. The bar chart highlights the differences between the original features that led to a 'Moderate' adaptability prediction and the Counterfactual features that would result in a 'Low' adaptability prediction. The original feature values are denoted in blue, while the Counterfactual values that could potentially alter the model's outcome are shown in orange. Notably, most features remain unchanged between the original and Counterfactual scenarios, except for 'Class Duration', which shows a decrement of 0.5 in the Counterfactual case. This decrease suggests that a reduction in 'Class Duration' is influential in shifting the adaptability prediction from 'Moderate' to 'Low'.

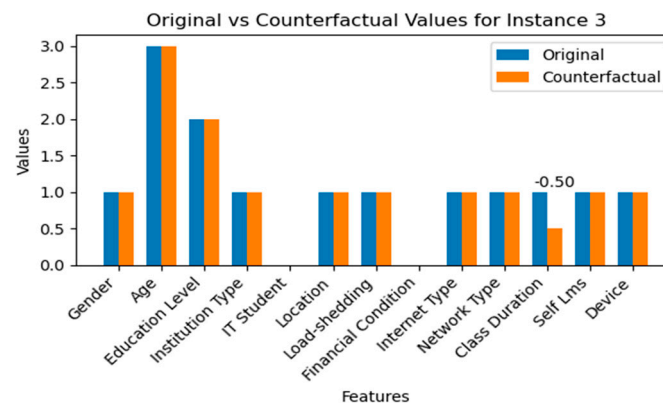


Figure 10. Comparative analysis of original and Counterfactual feature values for Instance 3.

The result also emphasizes the significant role that 'Class Duration' plays in the adaptability model, where a slight reduction in duration is the single change needed to alter the prediction. This finding has important implications for educational strategies, suggesting that maintaining or increasing class duration could be a critical factor in supporting higher adaptability levels among students. Educators and administrators may use these insights to reassess the structure of the school day and the allocation of time to different subjects. The chart encourages a data-driven approach to curriculum planning, ensuring that students have sufficient time for in-depth exploration and learning, which is crucial for developing adaptability skills. This Counterfactual analysis complements other interpretability techniques, such as SHAP and LIME, by providing a direct and tangible scenario where altering a single feature can change a student's predicted adaptability level. Such information is invaluable for policy formulation, as it highlights specific leverage points where educational interventions could be targeted to yield the most significant impact on student outcomes.

4.4. Comparison with Previous Studies

This study expands on earlier research by Hasan Suzan et al. [1], who predicted students' degrees of adaptation throughout the pandemic using the same dataset. By utilizing an extensive range of Explainable AI (XAI) methodologies, our research expands on previous findings and offers a more profound understanding of factors affecting adaptability.

Suzan et al. [1] used a variety of machine learning algorithms in their study, including Random Forest, Decision Tree, Naive Bayes, Support Vector Machine, K-Nearest Neighbors, and Artificial Neural Network. The Random Forest classifier yielded the best results, with an accuracy of 89.63% and precision values of 0.89, 0.90, 0.88 for low, moderate and high adaptability, respectively. Using the Random Forest model, our study, in contrast, obtained a higher accuracy of 91% and precision values of 0.93, 0.94, and 0.94 for high, low, and moderate adaptation levels, respectively. To provide a more thorough interpretation of the model predictions, we also used XAI techniques including SHAP, LIME, Anchors, ALE, and Counterfactual explanations to highlight certain feature contributions. We demonstrate the complementary nature of our approach and verify the robustness of our technique by comparing our results with those of [1]. Our work presents a more thorough knowledge of

student adaptation in the context of education, not only validating their findings but also providing improved interpretability using XAI approaches.

This comparison underscores the value of integrating XAI methods with traditional machine learning models to achieve both high predictive performance and interpretability, thereby contributing to more informed and effective educational interventions.

5. Comparative Analysis of XAI Techniques and Synthesis of Findings

After completing an extensive training program involving a varied set of machine learning models, our research delved into a thorough examination of model predictions using a range of XAI techniques. Each technique offered unique insights into the decision-making processes underlying the adaptability predictions, with SHAP and ALE providing a global view of feature contributions, LIME and Anchors elucidating local, instance-specific explanations, Counterfactuals revealing the sensitivity of predictions to feature perturbations, and ALE plots assessing the average feature effects across the data distribution. A comparative analysis of the XAI outputs revealed both similarities and divergence in feature importance rankings and explanations. Accordance across methods reinforced the perceived significance of certain features, such as ‘class duration’ and ‘financial condition’, suggesting these are robust determinants of adaptability predictions. Divergences prompted a careful reconsideration of complex feature interactions and model behavior in various scenarios, guiding further analytical probing.

This study leverages a diverse set of XAI techniques, including SHAP, LIME, Anchors, Counterfactual, and ALE, to comprehensively interpret student adaptability prediction models. The amalgamation of these methods sheds light on key features influencing adaptability levels, contextualizing predictions within the educational landscape. This holistic understanding forms the foundation for potential educational interventions, emphasizing the necessity of considering engagement duration and addressing socioeconomic factors.

The research makes a unique and valuable contribution to the areas of machine learning interpretability and educational policy-making by utilizing a combination of different XAI approaches to clarify the decision-making processes of educational data models. It not only enhances the interpretability of educational data through a systematic comparative analysis of various XAI techniques but also identifies and examines the complex socioeconomic, institutional, and demographic factors influencing student adaptability. The thorough exploration and results of this study provide the basis for developing detailed and targeted interventions. A notable feature of this research is its innovative methodology, which integrates and compares insights from several XAI techniques to gain a comprehensive understanding of model predictions. This method offers a detailed perspective on how various factors affect student adaptability, enabling the creation of well-informed and impactful educational policies and practices. By bridging the theoretical aspects of machine learning with practical, actionable insights, the study demonstrates the value of XAI in crafting educational strategies that are both equitable and tailored to enhance student outcomes. These contributions represent a significant advancement in the responsible application of AI within the educational sector, ensuring that predictive models are leveraged in a manner that is both transparent and beneficial to student learning and adaptability.

Limitations

Despite the remarkable findings and contributions of this study, it is important to acknowledge a number of limitations.

1. The dataset included in this study was sourced from a publicly accessible website (www.kaggle.com, accessed on 18 November 2023), despite being reliable and representative of a range of student characteristics. This restricts the extent of generalizability since the dataset might not include all the subtle characteristics seen in various educational environments throughout the world. Although the dataset effectively captures various demographic and socio-economic factors, it lacks contextual elements like cultural and extracurricular influences, which are also significant for a holistic

understanding of student adaptability. Despite these limitations, the dataset remains a valuable resource for exploring key factors influencing student adaptability and developing targeted educational interventions.

2. For analysis, the study mostly employed quantitative data. Qualitative data, from surveys or interviews with students, may provide deeper, more individualized insights into the variables influencing students' adaptation. This might aid in comprehending the contextual elements that are hidden by quantitative data on its own.
3. Cross-sectional data were used in the study which captures a snapshot of student adaptability at a single point in time. This method ignores periodic shifts in students' adaptability, which might offer more detailed information about how adaptability develops over time and is impacted by different kinds of interventions.

6. Conclusions

This study uses XAI techniques to understand factors influencing student adaptability in education. The application of various methods, such as SHAP, LIME, Anchors, ALE, and Counterfactual explanations, provides a comprehensive understanding of the dynamics influencing student adaptability. Key factors such as 'Class Duration', 'Financial Condition', 'Age', and 'Institution Type' are crucial in shaping students' ability to adapt in educational settings. These findings emphasize the importance of considering socio-economic and institutional factors when formulating educational strategies. XAI has the potential to foster a data-informed educational landscape, allowing for personalized learning pathways and adaptive educational systems. Integrating AI and analytics into educational policy and practice can enhance educational outcomes and create a more inclusive and effective learning environment.

Future inquiries should focus on analyzing longitudinal data to capture the temporal changes in adaptability and integrating qualitative data for richer, more personalized insights. Longitudinal studies would allow researchers to observe how adaptability evolves over time, providing a more dynamic understanding of the factors that influence it. Incorporating qualitative data, such as student interviews and surveys, can offer a more holistic view of student experiences and the contextual factors that quantitative data alone cannot reveal. This approach would enhance the depth of the analysis and potentially uncover new dimensions of student adaptability.

Evaluating the real-world effects of interventions informed by XAI and updating the interpretability framework to align with new machine learning advancements will be crucial. Ensuring the ethical application of AI and maintaining transparency in AI-driven educational tools remain essential for the responsible progression of this field.

We suggest adding a validation stage to the model creation procedure for future studies. This innovation may make it easier to fine-tune and validate the model in more subtle ways, which might improve interpretability and prediction performance. Investigating how a certain validation set affects the generalizability of machine learning models in educational contexts may yield insightful information that advances our knowledge of student flexibility and the efficient use of XAI methods.

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